In [1]:
```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [3]:
```python
df = sns.load_dataset('iris')
```

In [4]:
```python
print("Shape:", df.shape)
print("\nColumns:", df.columns)
print("\nFirst 5 rows:\n", df.head())
```

```
Shape: (150, 5)

Columns: Index(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
       'species'],
      dtype='object')

First 5 rows:
    sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa
```

In [5]:
```python
df.info()
```

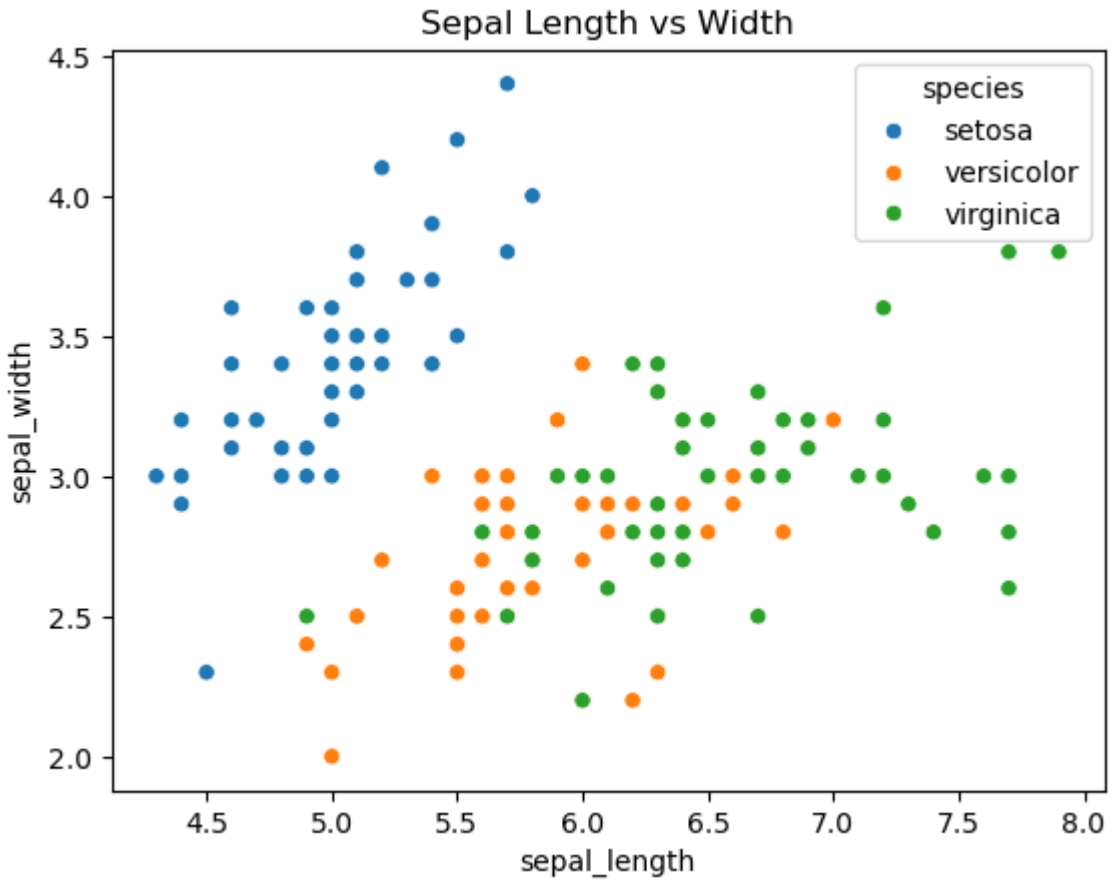
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

In [6]:
```
df.describe()
```

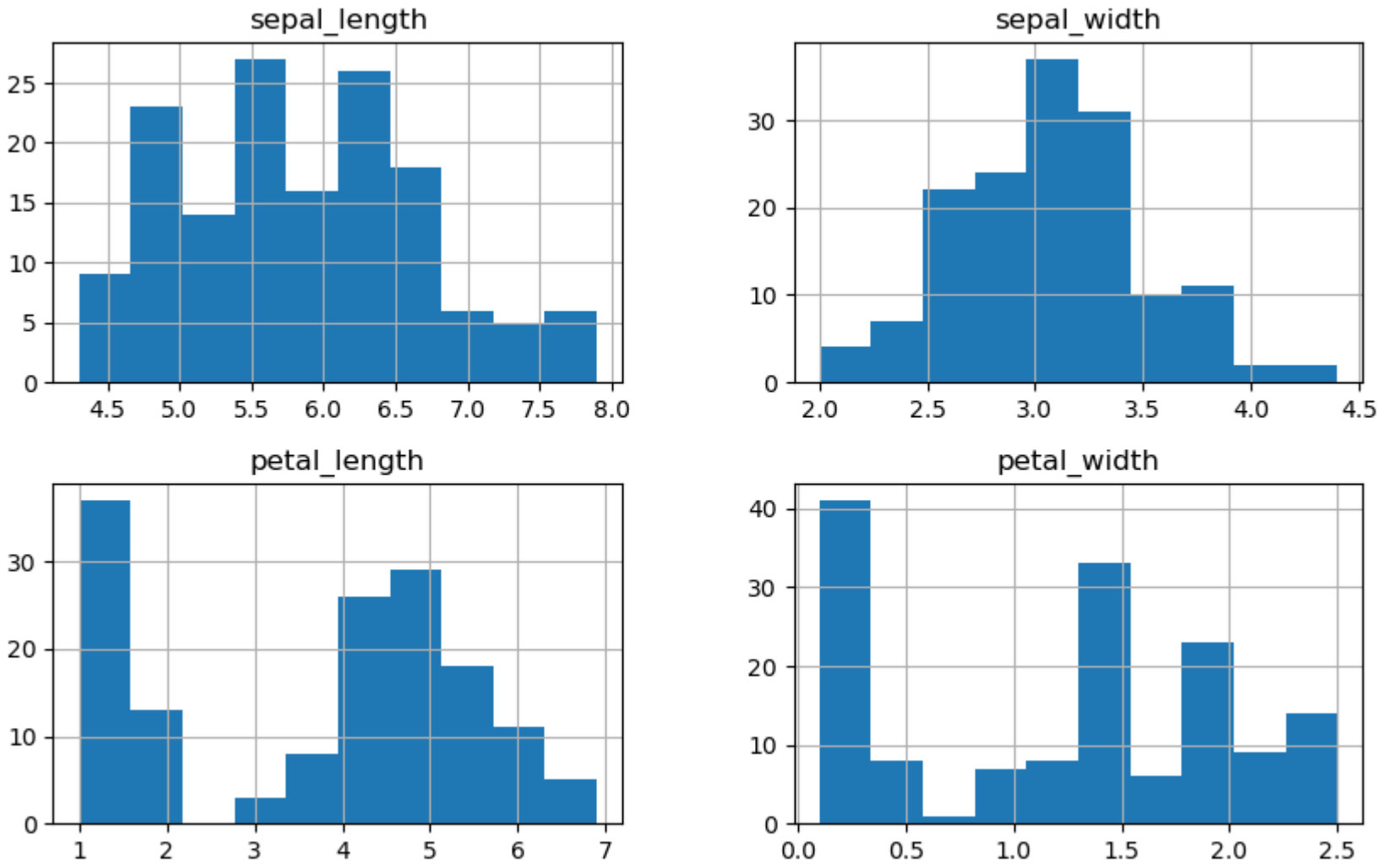
Out[6]:

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

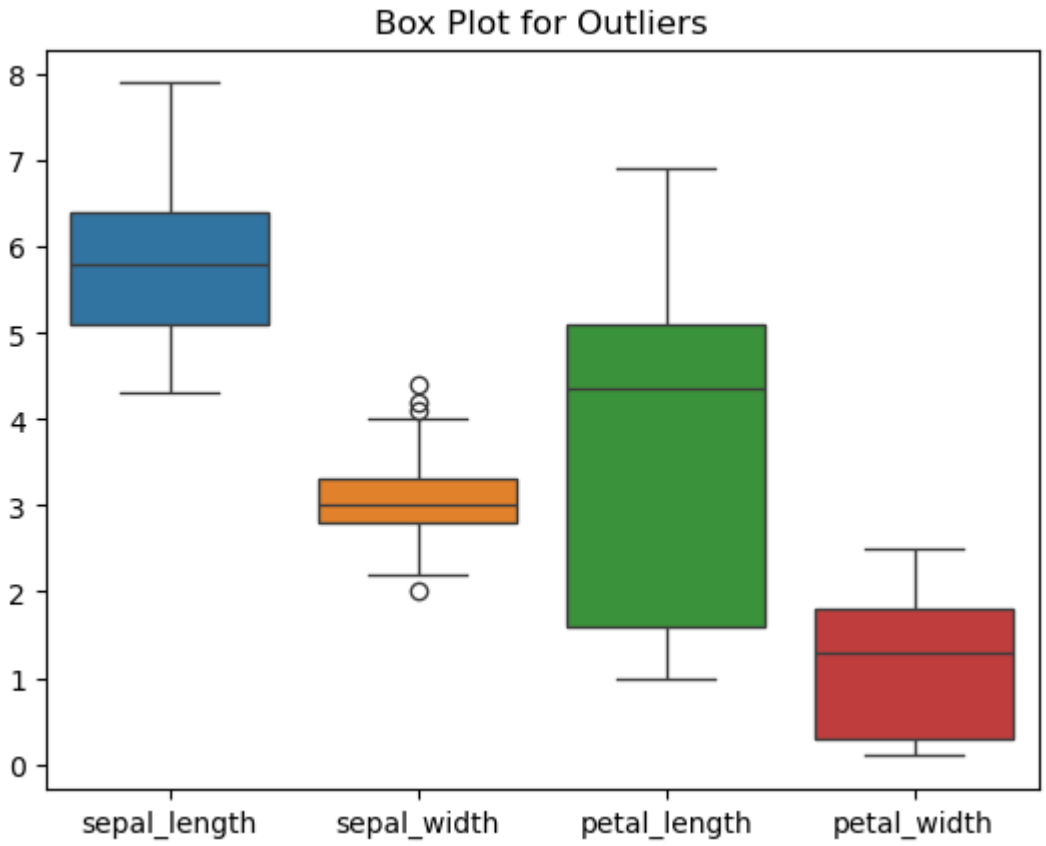
In [7]:
```
sns.scatterplot(x='sepal_length', y='sepal_width', hue='species', data=df)
plt.title("Sepal Length vs Width")
plt.show()
```



In [8]:
```
df.hist(figsize=(10,6))
plt.show()
```



In [9]:
```
sns.boxplot(data=df)
plt.title("Box Plot for Outliers")
plt.show()
```



In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")

sns.set_style("whitegrid")
plt.rcParams["figure.figsize"] = (10, 6)
print("Libraries imported successfully!")
print(f"pandas  version: {pd.__version__}")
print(f"seaborn version: {sns.__version__}")
```

```
Libraries imported successfully!
pandas  version: 2.3.3
seaborn version: 0.13.2
```

In [2]:

```python
# Load Iris dataset using seaborn (no download needed)
df = sns.load_dataset("iris")

print("Dataset loaded successfully!")
print(f"Shape         : {df.shape}  (rows x columns)")
print(f"Column names  : {list(df.columns)}")
print()
print("=== First 10 Rows ===")
df.head(10)
```

```
Dataset loaded successfully!
Shape         : (150, 5)  (rows x columns)
Column names  : ['sepal_length', 'sepal_width', 'petal_length', 'petal_width', 'species']

=== First 10 Rows ===
```

Out[2]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
5	5.4	3.9	1.7	0.4	setosa
6	4.6	3.4	1.4	0.3	setosa
7	5.0	3.4	1.5	0.2	setosa
8	4.4	2.9	1.4	0.2	setosa
9	4.9	3.1	1.5	0.1	setosa

In [3]:
```python
print("=== .info() — Data Types & Memory ===")
df.info()
print()
print("=== Missing Values per Column ===")
print(df.isnull().sum())
print()
print("=== Species Class Distribution ===")
print(df["species"].value_counts())
```

```
=== .info() — Data Types & Memory ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB

=== Missing Values per Column ===
sepal_length    0
sepal_width     0
petal_length    0
petal_width     0
species         0
dtype: int64

=== Species Class Distribution ===
species
setosa        50
versicolor    50
virginica     50
Name: count, dtype: int64
```

In [4]:
```python
print("=== .describe() — Descriptive Statistics ===")
df.describe()
```

```
=== .describe() — Descriptive Statistics ===
```

Out[4]:

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

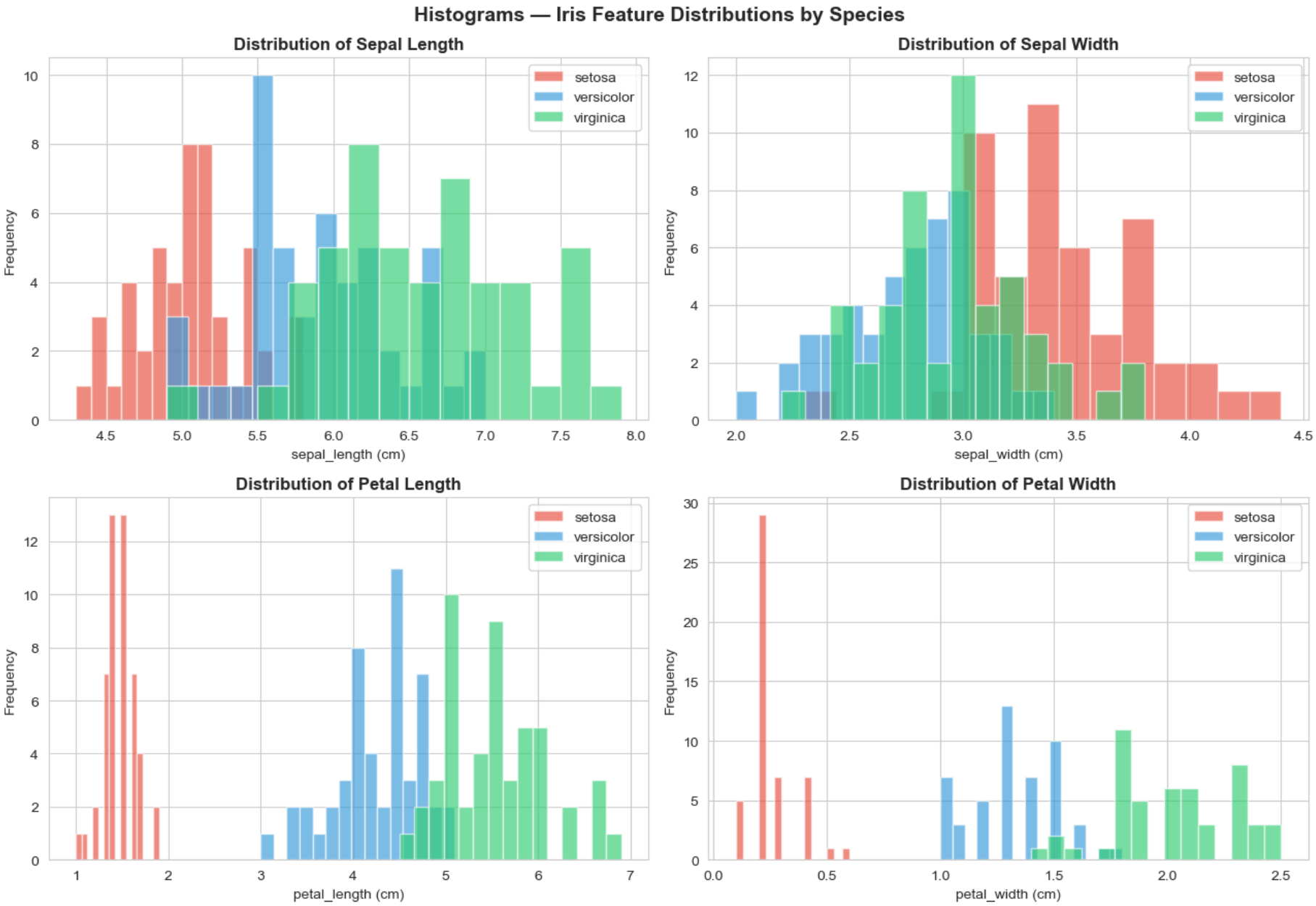
In [5]:

```python
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Plot 1: Sepal dimensions
sns.scatterplot(ax=axes[0], data=df, x="sepal_length", y="sepal_width",
                hue="species", palette="Set1", s=80, edgecolor="white")
axes[0].set_title("Sepal Length vs Sepal Width", fontsize=13, fontweight="bold")
axes[0].set_xlabel("Sepal Length (cm)")
axes[0].set_ylabel("Sepal Width (cm)")

# Plot 2: Petal dimensions
sns.scatterplot(ax=axes[1], data=df, x="petal_length", y="petal_width",
                hue="species", palette="Set1", s=80, edgecolor="white")
axes[1].set_title("Petal Length vs Petal Width", fontsize=13, fontweight="bold")
axes[1].set_xlabel("Petal Length (cm)")
axes[1].set_ylabel("Petal Width (cm)")

plt.suptitle("Iris Dataset — Scatter Plots by Species", fontsize=15, fontweight="bold")
plt.tight_layout()
plt.savefig("scatter_plots.png", dpi=100, bbox_inches="tight")
plt.show()
print("Observation: Petal features clearly separate species better than sepal features!")
```

Observation: Petal features clearly separate species better than sepal features!

In [6]:
```
features = ["sepal_length", "sepal_width", "petal_length", "petal_width"]
species_colors = {"setosa": "#e74c3c", "versicolor": "#3498db", "virginica": "#2ecc71"}

fig, axes = plt.subplots(2, 2, figsize=(13, 9))
axes = axes.flatten()

for i, feat in enumerate(features):
    for species, color in species_colors.items():
        subset = df[df["species"] == species][feat]
        axes[i].hist(subset, bins=15, alpha=0.65, color=color,
                     label=species, edgecolor="white")
    axes[i].set_title(f"Distribution of {feat.replace('_', ' ').title()}",
                      fontsize=12, fontweight="bold")
    axes[i].set_xlabel(f"{feat} (cm)")
    axes[i].set_ylabel("Frequency")
    axes[i].legend()

plt.suptitle("Histograms — Iris Feature Distributions by Species",
             fontsize=14, fontweight="bold")
```

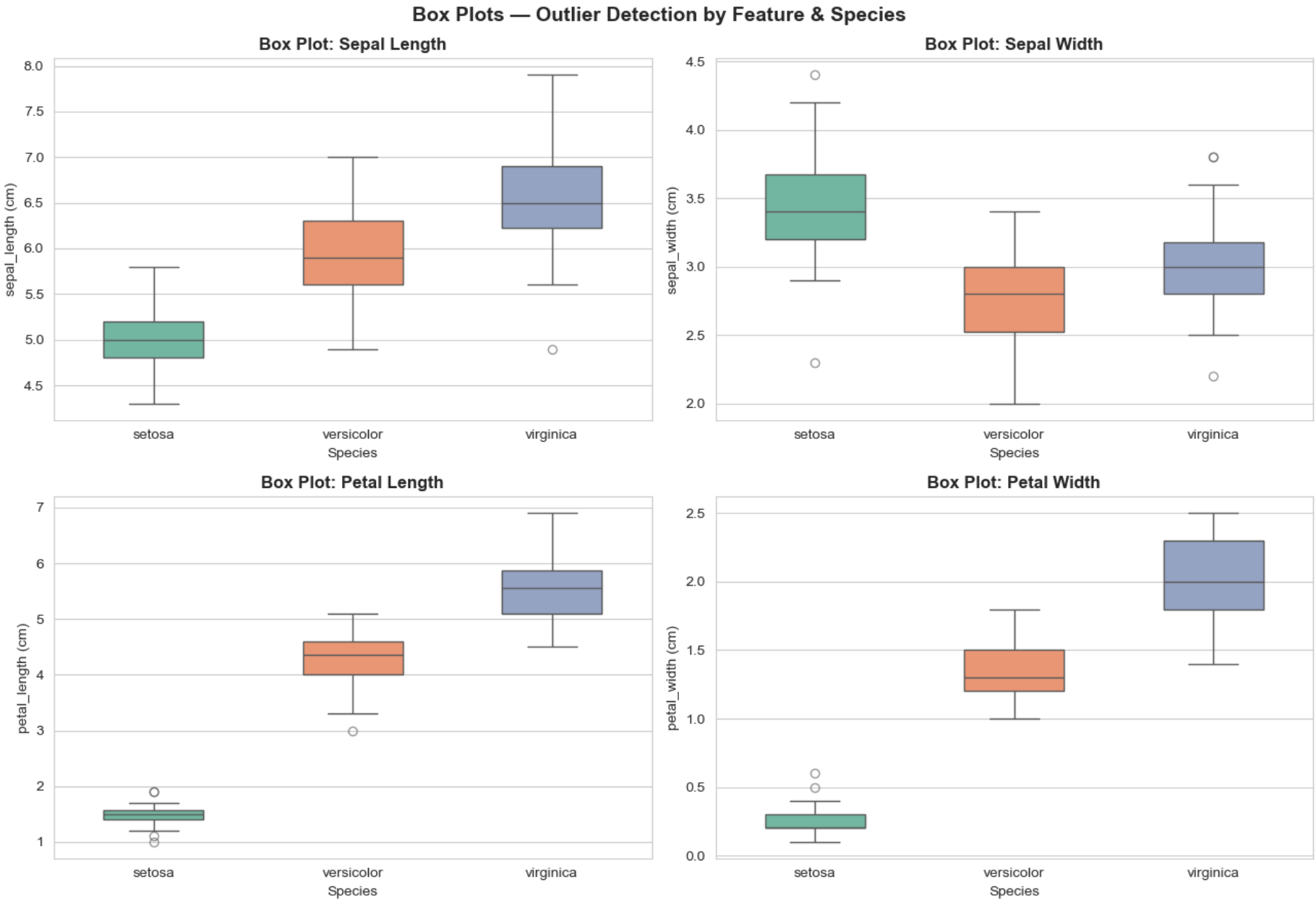
```python
plt.tight_layout()
plt.savefig("histograms.png", dpi=100, bbox_inches="tight")
plt.show()
print("Observation: Petal length/width shows the clearest species separation.")
```



Observation: Petal length/width shows the clearest species separation.

In [7]:
```
fig, axes = plt.subplots(2, 2, figsize=(13, 9))
```

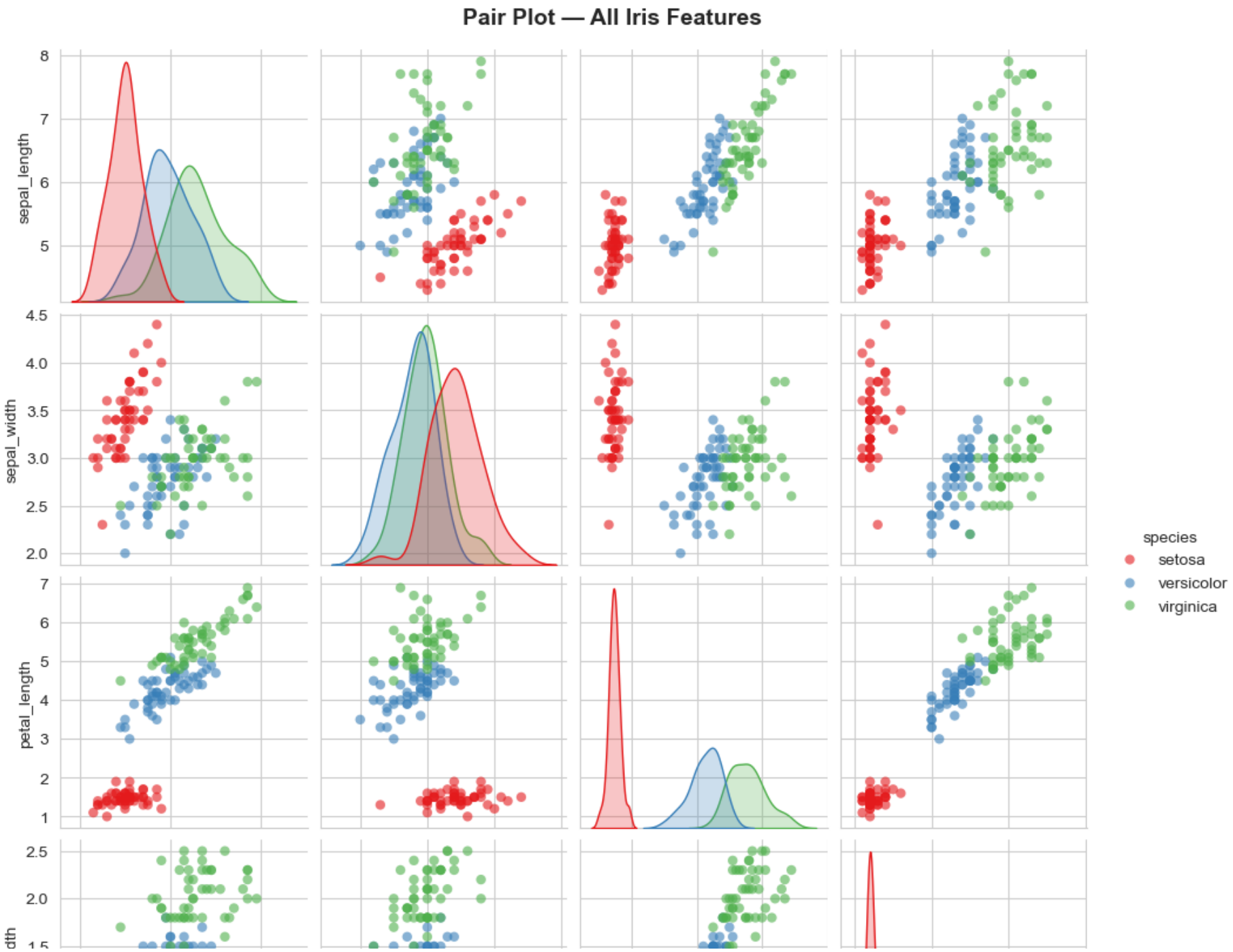
```python
axes = axes.flatten()

for i, feat in enumerate(features):
    sns.boxplot(ax=axes[i], data=df, x="species", y=feat,
                palette="Set2", width=0.5,
                flierprops=dict(marker="o", color="red", markersize=6, alpha=0.6))
    axes[i].set_title(f"Box Plot: {feat.replace('_', ' ').title()}",
                      fontsize=12, fontweight="bold")
    axes[i].set_xlabel("Species")
    axes[i].set_ylabel(f"{feat} (cm)")

plt.suptitle("Box Plots — Outlier Detection by Feature & Species",
             fontsize=14, fontweight="bold")
plt.tight_layout()
plt.savefig("boxplots.png", dpi=100, bbox_inches="tight")
plt.show()
print("Observation: Setosa has notably smaller petal dimensions. Some sepal_width outliers visible.")
```

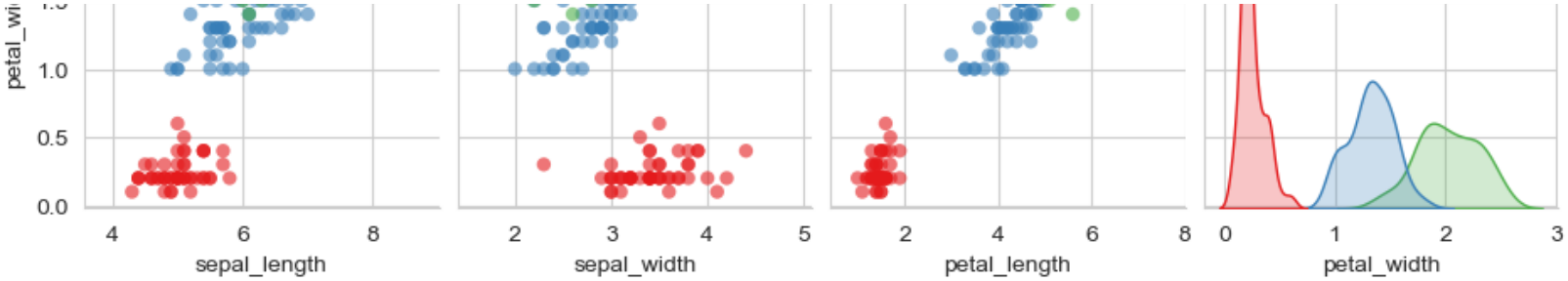


Observation: Setosa has notably smaller petal dimensions. Some sepal_width outliers visible.

In [8]:
```
pair = sns.pairplot(df, hue="species", palette="Set1",
```

```
                     diag_kind="kde", height=2.4,
                     plot_kws={"alpha": 0.6, "edgecolor": "none"})
pair.fig.suptitle("Pair Plot — All Iris Features", y=1.02,
                   fontsize=14, fontweight="bold")
plt.savefig("pairplot.png", dpi=100, bbox_inches="tight")
plt.show()
print("Pair plot saved — shows all pairwise relationships!")
```

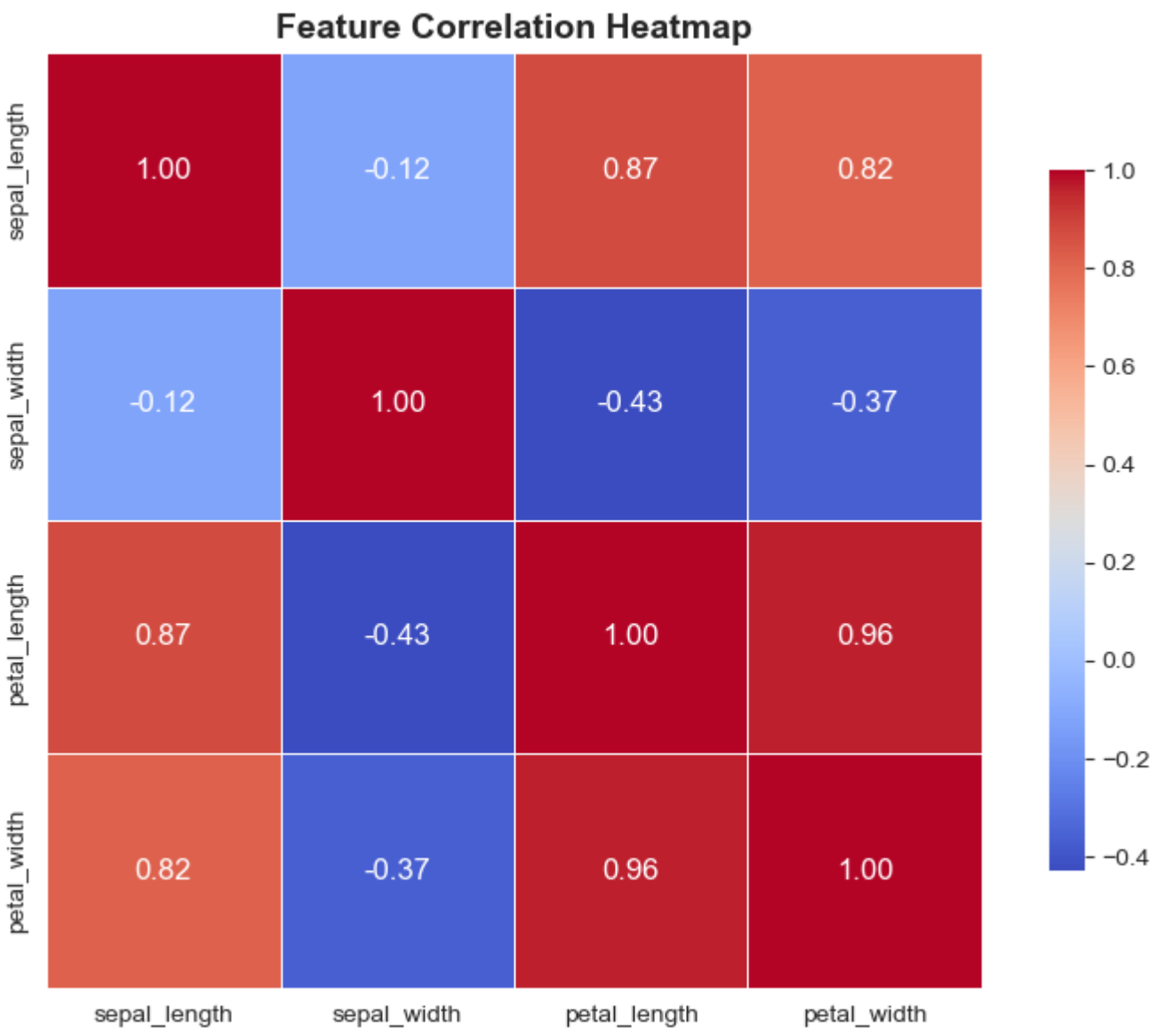

Pair Plot — All Iris Features
sepal_length
sepal_width
petal_length
species
setosa
versicolor
virginica




```
Pair plot saved — shows all pairwise relationships!
```

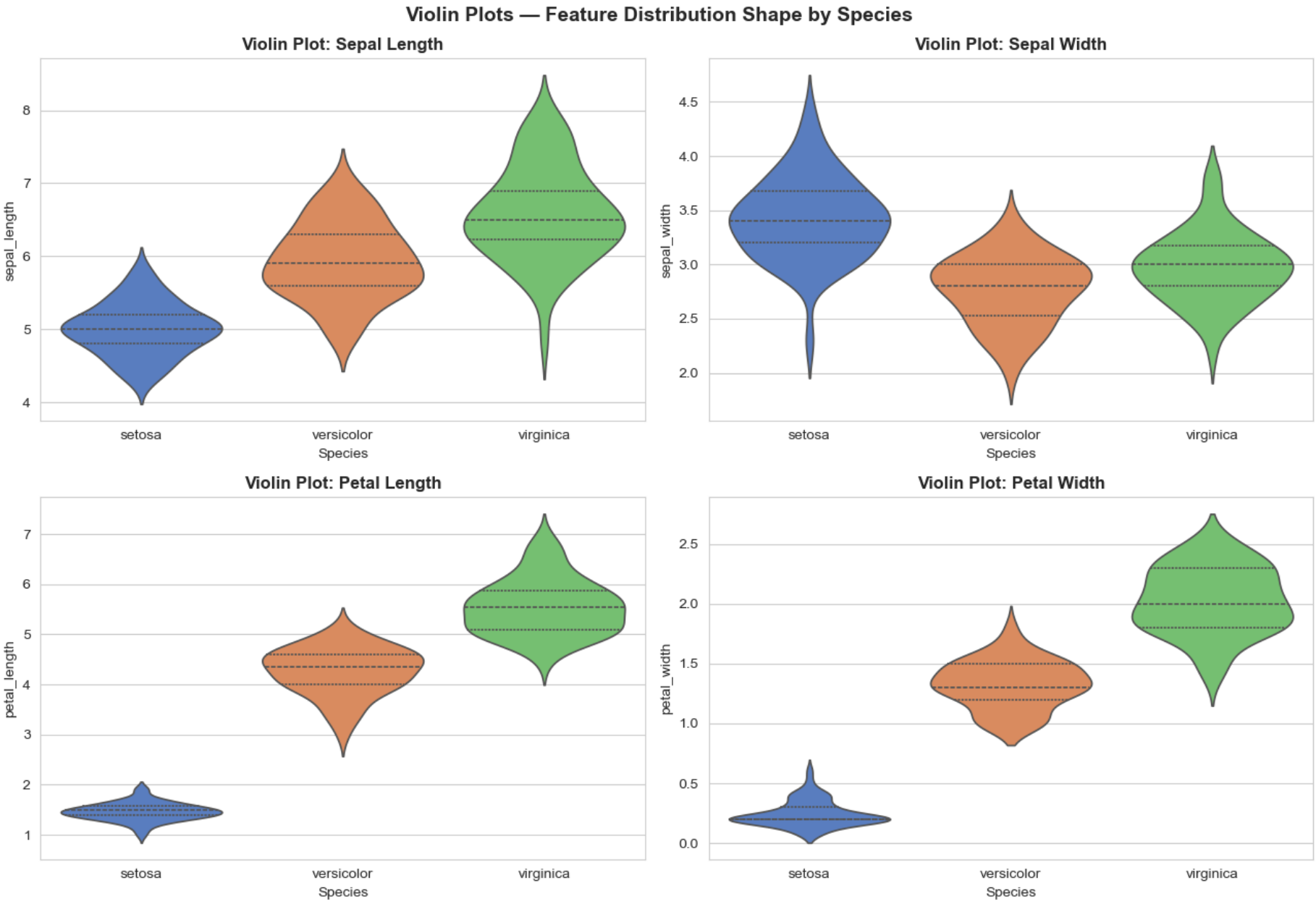
In [9]:

```
plt.figure(figsize=(8, 6))
corr = df[features].corr()
sns.heatmap(corr, annot=True, fmt=".2f", cmap="coolwarm",
            linewidths=0.5, square=True,
            cbar_kws={"shrink": 0.75},
            annot_kws={"size": 12})
plt.title("Feature Correlation Heatmap", fontsize=14, fontweight="bold")
plt.tight_layout()
plt.savefig("heatmap.png", dpi=100, bbox_inches="tight")
plt.show()
```



In [10]:
```
fig, axes = plt.subplots(2, 2, figsize=(13, 9))
axes = axes.flatten()

for i, feat in enumerate(features):
    sns.violinplot(ax=axes[i], data=df, x="species", y=feat,
```

```
                    palette="muted", inner="quartile")
    axes[i].set_title(f"Violin Plot: {feat.replace('_', ' ').title()}",
                      fontsize=12, fontweight="bold")
    axes[i].set_xlabel("Species")

plt.suptitle("Violin Plots — Feature Distribution Shape by Species",
             fontsize=14, fontweight="bold")
plt.tight_layout()
plt.savefig("violin_plots.png", dpi=100, bbox_inches="tight")
plt.show()
```



In [11]:
```
print("=== Summary Statistics by Species ===")
print()
```

```python
summary = df.groupby("species")[features].agg(["mean", "std", "min", "max"])
print(summary.round(2))
```

```
=== Summary Statistics by Species ===

           sepal_length                 sepal_width                  \
                   mean   std  min  max        mean   std  min  max
species
setosa             5.01  0.35  4.3  5.8        3.43  0.38  2.3  4.4
versicolor         5.94  0.52  4.9  7.0        2.77  0.31  2.0  3.4
virginica          6.59  0.64  4.9  7.9        2.97  0.32  2.2  3.8

           petal_length                 petal_width
                   mean   std  min  max        mean   std  min  max
species
setosa             1.46  0.17  1.0  1.9        0.25  0.11  0.1  0.6
versicolor         4.26  0.47  3.0  5.1        1.33  0.20  1.0  1.8
virginica          5.55  0.55  4.5  6.9        2.03  0.27  1.4  2.5
```

In [12]:

```python
print("="*60)
print("KEY INSIGHTS — IRIS DATASET EDA")
print("="*60)
print()
print("1. DATASET OVERVIEW:")
print(f"   150 samples | 4 numeric features | 3 species (50 each)")
print(f"   Zero missing values — clean dataset!")
print()
print("2. FEATURE SEPARABILITY:")
print("   Petal length & width = BEST for species classification")
print("   Sepal features show more overlap between species")
print()
print("3. SPECIES CHARACTERISTICS:")
print("   Setosa  : smallest petals (very distinct from others)")
print("   Versicolor: medium-sized features")
print("   Virginica : largest petals and sepal length")
print()
print("4. CORRELATIONS (from heatmap):")
corr = df[features].corr()
print(f"   Petal length vs Petal width : {corr.loc['petal_length','petal_width']:.2f}  (very high!)")
print(f"   Petal length vs Sepal length: {corr.loc['petal_length','sepal_length']:.2f}  (high)")
print(f"   Sepal width  vs Petal length: {corr.loc['sepal_width','petal_length']:.2f}  (negative)")
```

```python
print()
print("5. OUTLIERS:")
print("   Minor outliers in sepal_width for Setosa")
print("   No extreme outliers affecting overall analysis")
print()
print("CONCLUSION:")
print("   Iris dataset is ideal for classification ML tasks.")
print("   Petal features alone can achieve near-perfect classification.")
```

```
============================================================
KEY INSIGHTS — IRIS DATASET EDA
============================================================

1. DATASET OVERVIEW:
   150 samples | 4 numeric features | 3 species (50 each)
   Zero missing values — clean dataset!

2. FEATURE SEPARABILITY:
   Petal length & width = BEST for species classification
   Sepal features show more overlap between species

3. SPECIES CHARACTERISTICS:
   Setosa  : smallest petals (very distinct from others)
   Versicolor: medium-sized features
   Virginica : largest petals and sepal length

4. CORRELATIONS (from heatmap):
   Petal length vs Petal width : 0.96  (very high!)
   Petal length vs Sepal length: 0.87  (high)
   Sepal width  vs Petal length: -0.43  (negative)

5. OUTLIERS:
   Minor outliers in sepal_width for Setosa
   No extreme outliers affecting overall analysis

CONCLUSION:
   Iris dataset is ideal for classification ML tasks.
   Petal features alone can achieve near-perfect classification.
```

In [ ]: